higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER PROGRAMMING
(Second Paper)
NQF LEVEL 4

14 MARCH 2019

This marking guideline consists of 15 pages.

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COMPLETE ANSWERS ARE INCLUDED AT THE END OF THE MARKING GUIDELINE.

QUESTION 1

1.1 1.1.1

```
Public Class Evaluate
  Private strMake As String ✓
  Private purchasePrice As Double 'make any two variables
  Private Mileage As Integer
  Private percent As Decimal ✓
```

(2)

1.1.2

```
Public Property make() As String
        Get
            Return strMake
        End Get
        Set(ByVal value As String)
            strMake = value
        End Set
    End Property
    Public Property price() As Double
        Get
            Return purchasePrice         End Get
        Set(ByVal value As Double)

            purchasePrice = value
        End Set
    End Property
    Public Property mile() As Integer✓
        Get
            Return Mileage ✓
        End Get
        Set(ByVal value As Integer)
            Select Case value ✓
                Case 10000 To 39999 ✓
                    percent = 0.15
                Case 40000 To 99999✓
                    percent = 0.35
                Case 100000 To 199999✓
                    percent = 0.65
                Case 200000 To 400000
                    percent = 0.75 ✓
            End Select‘

            Mileage = value
        End Set
    End Property
```

(7)

1.1.3

```
    Public ReadOnly✓ Property percentage() As Decimal '✓
        Get
            Return percent ✓
        End Get
```

(3)

1.1.4
```
Public Function calResale()
Dim resalePrice As Double
resalePrice = purchasePrice ✓- (purchasePrice *
percentage✓)
Return FormatCurrency(resalePrice) ✓✓
End Function
End Class
```
(4)

1.2 1.2.1
```
Public Class frmCarEvaluate
    Private selectMake As String
    Protected purchasePrice As Double
    Protected mileage As Integer
  Protected resellPrice As Double
Dim objEvaluate As New clsCarEvaluator ✓✓
```
(2)

1.2.2
```
Private Sub btnEvaluate_Click(...) Handles
btnEvaluate.Click
          objEvaluate.make = cmbMake.SelectedItem✓
        objEvaluate.price = txtpurchase.Text ✓
        objEvaluate.mile = txtmileage.Text ✓
```
(3)

1.2.3
```
lblmake.Text = objEvaluate.make
lblpurchase.Text = FormatCurrency(objEvaluate.price) ✓
lblmiileage.Text = objEvaluate.mile
lblprice.Text = objEvaluate.calResale()✓
```
(2)

1.3 Correct form output printout ✓ (2)
[25]

QUESTION 2

2.1 2.1.1
```
Sub getBooking()✓
        Try
            connection.ConnectionString = ✓
"Provider=Microsoft.ACE.OLEDB.12.0;Data Source=C:\Users
\CBS_Systems.accdb"

            connection.Open() ✓       only this 3 section
```
(3)

2.1.2
```
myquery.Connection = connection
myquery.CommandText = "INSERT INTO ✓ BookAssessment✓
(AssessID,AssessmentDate,studentID)
VALUES(@assessID,@assessDate,@id)" ✓
 Dim assess = New OleDb.OleDbParameter("@assessid",
cmbAssessment.SelectedItem) ✓
 Dim sdate = New OleDb.OleDbParameter("@assessDate",
dtpDate.Text) ✓
 Dim sidno = New OleDb.OleDbParameter("@id",
txtsearchIdno.Text) ✓
            myquery.Parameters.Add(assess) ✓
            myquery.Parameters.Add(sdate) ✓
            myquery.Parameters.Add(sidno) ✓
            myquery.ExecuteNonQuery()✓
            connection.Close()
        Catch ex As Exception
            MessageBox.Show(ex.ToString)
        End Try
```
(10)

2.2
```
Private Sub btnbooking_Click(…) Handles btnbooking.Click
        getBooking()✓
    End Sub
```
(1)

2.3
```
Sub getSchedule()
        Try
            connection.ConnectionString =
"Provider=Microsoft.ACE.OLEDB.12.0;Data Source=C:\Users\Session1.A4-
PC13\Downloads\CBS_Systems.accdb"
            connection.Open()
            myquery.Connection = connection
            myquery.CommandText = "SELECT ✓ Register.IdentityNr,
Register.LastName, Assessment.AssessmentName,
BookAssessment.AssessmentDate FROM ✓Register INNER JOIN (Assessment
INNER JOIN BookAssessment ON Assessment.AssessmentID =
✓BookAssessment.AssessID) ON Register.IdentityNr =
BookAssessment.StudentID WHERE ((BookAssessment.StudentID =@studID
✓))"
Dim stuID = New OleDb.OleDbParameter("StudID", txtsearch.Text) ✓
            myquery.Parameters.Add(stuID) ✓
            table.Load(myquery.ExecuteReader) ✓
```
(7)

2.4
```
dgvSchedule.DataSource = table✓
        Catch ex As Exception
            MessageBox.Show(ex.ToString)
        End Try
    End Sub
```
(1)

2.5
```
Private Sub btnSearch_Click(…) Handles btnSearch.Click
        getSchedule()✓
    End Sub
```
(1)

2.6 Correct form output printout ✓✓ (2)

[25]

QUESTION 3

3.1 3.1.1
```
#navigation✓ { {height: 20px; width: 570px; text-align:center;
padding:0px 15px;}
✓
```
(2)

3.1.2
```
#rightside { ✓
position:absolute;left:320px; right:20px;width: 150px;✓
padding: 5px 15px;
}
```
(2)

3.1.3
```
a { text-decoration: none; ✓}
```
(1)

3.1.4
```
a:hover ✓{background-color:gray; ✓}
```
(2)

3.1.5
```
#footer {
clear: both;
background-color:silver;
text-align:center; ✓

}
```
(1)

3.2 3.2.1
```
<html>
<head>
<title>WBE Survey</title>
<link rel="stylesheet"href="style1.css"type="text/css">
</head>
<body>
   <div id="container">✓
   <div id="header"> ✓
```
(2)

3.2.2
```
<h1> ✓ WBE SURVEY FOR TVET COLLEGES!</h1>
   </div>
```
(1)

3.2.3
```
<div id="navigation"> ✓
✓ <a href="TVETColleges.html ">TVET Colleges</a>
<a href="Sponsorship.html ">Sponsorship </a>
<a href="ApplicationForm.html">Application form</a> ✓
   </div>
```
(3)

3.2.4
```
<div id='rightside'>✓
<img src='images/survey2.jpg✓ 'height='150px' width='230px'✓>
</div>
```
(3)

3.2.5
```
<div id="content">✓
<table border="0"> ✓
<form action="" method="post" onsubmit="return ()">
```
(2)

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3.2.6
```
<tr><td> ✓Surname:</td> <td><input type="text"
name="surname1" id="sname1" /></td></tr>
<tr><td>Email: </td> <td>  ✓<input type="text" name="email"
id="email" /></td></tr>

<tr><td> Specialization:<br></td> <td>
 ✓ <INPUT TYPE="radio" NAME="rad" Value="Networking"
>Networking <BR>
<INPUT TYPE="radio" NAME="rad" Value="Programming"
>Programming<BR>
✓  <INPUT  TYPE="radio"  NAME="rad"  Value="System
Analysis">Programming<BR></td></tr>
```
(6)

3.2.7
```
<span id="cty">
<tr><td>Province: ✓</td> <td><select name="province"
✓id="province">
<option value="">Choose a province</option>✓
<option value="ec">Eastern Cape</option>✓
<option value="gp">Gauteng</option>✓
<option value="fs">Free State</option>
<option value="kzn">Kwazulu Natal</option>
<option value="lp">Limpopo</option>
<option value="mp">Mpumalanga</option>
<option value="nc">Northern Cape</option>
<option value="nw">North West</option>
<option value="wc">Western Cape</option></td></tr>
</select>
```
(5)

3.2.8
```
<tr><td></td><td><input type="submit" ✓value="Submit" />
   
<input type="reset" ✓ value="Clear" /></td></tr>
</table></form>
</div>
```
(2)

3.2.9
```
<div id="footer">
      YOUR EXAMINATION MUMBER  ✓
   </div>
</div>
</body
</html>
```
(1)
[33]

QUESTION 4

4.1 4.1.1 <script type="text/javascript"> ✓ (1)

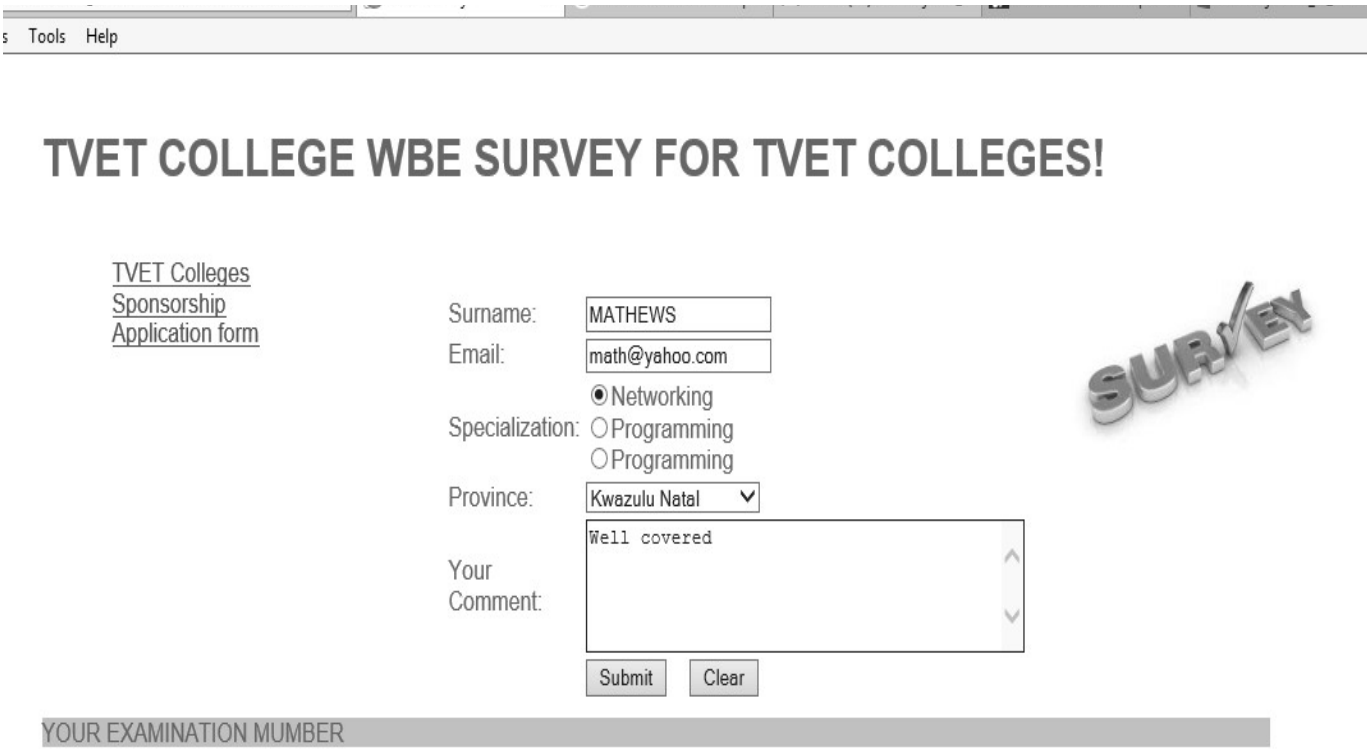
4.1.2
```
function survey()✓✓
{
var txt = document.getElementById("sname1");✓
var txte = document.getElementById("email");✓

if(txt.value== "" || txt.value== null) ✓
 {
alert("Please enter a name");
txt.style.border = "2px solid red";✓
txt.focus()✓
return false;
} else {
txt.style.border = "";
}
if(txte.value== "" || txte.value== null) ✓
 {
alert("Please enter an email");
txte.style.border = "2px solid red";
txte.focus()
return false;
} else {
txte.style.border = "";
}
```
(Allocate 4 to each text box validation) (8)

4.1.3
```
var combo1 = document.getElementById("province")✓✓
if ✓ (combo1.value == null || combo1.value == "") ✓ {
```
(4)

4.1.4
```
alert("Please select a province");✓
document.getElementById("cty").style.border = "2px solid red";✓
return false;
} else {✓
document.getElementById("cty").style.border = "";✓
}
}
</script>
```
(4)

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[17]

TOTAL: 100

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COMPLETE ANSWERS

QUESTION 1

```
Public Class clsCarEvaluator
    Private strMake As String
    Private purchasePrice As Double 'make any two variables
    Private Mileage As Integer
    Private percent As Decimal

    Public Property make() As String
        Get
            Return strMake
        End Get
        Set(ByVal value As String)
            strMake = value
        End Set
    End Property


    Public Property price() As Double
        Get
            Return purchasePrice
        End Get
        Set(ByVal value As Double)

            purchasePrice = value
        End Set
    End Property
    Public Property mile() As Integer
        Get
            Return Mileage
        End Get
        Set(ByVal value As Integer)
            Select Case value
                Case 10000 To 39999
                    percent = 0.15
                Case 40000 To 99999
                    percent = 0.35
                Case 100000 To 199999
                    percent = 0.65
                Case 200000 To 400000
                    percent = 0.75
            End Select
            Mileage = value
        End Set
    End Property
    Public ReadOnly Property percentage() As Decimal '
        Get
            Return percent
        End Get
    End Property
    Public Function calDepreciation()
        Dim resalePrice As Double
        resalePrice = purchasePrice - purchasePrice * percentage

        Return FormatCurrency(resalePrice)
    End Function
End Class
```

```
Public Class frmCarEvaluate
    Private selectMake As String
    Protected purchasePrice As Double
    Protected mileage As Integer
    Protected resellPrice As Double

    Private Sub btnEvaluate_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnEvaluate.Click
        Dim objEvaluate As New clsCarEvaluator
        objEvaluate.make = cmbMake.SelectedItem
        objEvaluate.price = txtPurchase.Text
        objEvaluate.mile = txtMileage.Text
        lblMake.Text = objEvaluate.make
        lblPurchase.Text = FormatCurrency(objEvaluate.price)
        lblMileage.Text = objEvaluate.mile
        lblPrice.Text = objEvaluate.calDepreciation()
    End Sub

    Private Sub btnClear_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnClear.Click
        cmbMake.SelectedIndex = 0
        txtPurchase.Clear()
        txtMileage.Clear()
        lblMake.Text = ""
        lblPurchase.Text = ""
        lblMileage.Text = ""
        lblPrice.Text = ""
    End Sub
End Class
```

QUESTION 2

```
Public Class CBT_Systems
    'global variables
    'declared to be used in the sub procedures
    Dim connection As New OleDb.OleDbConnection
    Dim myquery As New OleDb.OleDbCommand
    Dim table As New Data.DataTable
    ''Question 2.1      2.1.1
    Sub getBooking()
        Try
            connection.ConnectionString = "Provider=Microsoft.ACE.OLEDB.12.0;Data
Source=C:\Users \CBS_Systems.accdb"
            connection.Open()

            ''Question  2.1.2
            myquery.Connection = connection
            myquery.CommandText = "INSERT INTO BookAssessment
(AssessID,AssessmentDate,studentID) " & _
                                  "VALUES(@assessID,@assessDate,@id)"
            Dim assess = New OleDb.OleDbParameter("@assessid",
cmbAssessment.SelectedItem)
            Dim sdate = New OleDb.OleDbParameter("@assessDate", dtpDate.Text)
            Dim sidno = New OleDb.OleDbParameter("@id", txtsearchIdno.Text)
            myquery.Parameters.Add(assess)
            myquery.Parameters.Add(sdate)
            myquery.Parameters.Add(sidno)
            myquery.ExecuteNonQuery()
            connection.Close()
        Catch ex As Exception
            MessageBox.Show(ex.ToString)
```

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```
        End Try
    End Sub


    '' Question 2.4     2.4.1
    Sub getSchedule()
        Try
            connection.ConnectionString = "Provider=Microsoft.ACE.OLEDB.12.0;Data
Source=C:\Users\Session1.A4-PC13\Downloads\CBS_Systems.accdb"
            connection.Open()
            myquery.Connection = connection
            myquery.CommandText = "SELECT Register.IdentityNr, Register.LastName,
Assessment.AssessmentName, BookAssessment.AssessmentDate " & _
                                  "FROM Register INNER JOIN (Assessment " & _
                                  "INNER JOIN BookAssessment ON
Assessment.AssessmentID =  BookAssessment.AssessID) " & _
                                  "ON Register.IdentityNr =
BookAssessment.StudentID " & _
                                  "WHERE ((BookAssessment.StudentID =@studID  ))"
            Dim stuID = New OleDb.OleDbParameter("StudID", txtsearch.Text)
            myquery.Parameters.Add(stuID)
            table.Load(myquery.ExecuteReader)
            connection.Close()

            ''Question 2.5
            dgvSchedule.DataSource = table
        Catch ex As Exception
            MessageBox.Show(ex.ToString)
        End Try
    End Sub

    Private Sub btnAccount_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnAccount.Click
        tbcAccount.SelectedTab = tbpBooking
        'the tabcontrol opens the booking tabpga
    End Sub

    ''Question 2.3      2.3.1
    Private Sub btnbooking_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnBooking.Click
        getBooking()

    End Sub ''(1)
    Sub clearall()
        txtEmail.Clear()
        txtsearchIdno.Clear()
    End Sub

    Private Sub btnclose_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnClose.Click
        Application.Exit()
    End Sub

    Private Sub btnSearch_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnSearch.Click
        getSchedule()
    End Sub

    Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs)
        Application.Exit()
```

```
    End Sub


    Private Sub bntExit_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs)
        '' Exit application
        Application.Exit()
    End Sub

    Private Sub bntClear_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles bntClear.Click
        clearall()
    End Sub
End Class
```

QUESTION 3

```
<! Question 3.1  3.1.1>

body {
  font-family: 'Open Sans', sans-serif;
  color: #666;
}

#container {
padding: 5px;width: 750px;
margin: 20px auto;
position:absolute;left:120px; right:120px;
}

header {
     height: 100px;
     padding: 0 15px;
}
#content {
     width: 364px;
     float: left;
     padding: 5px 10px;
     margin: 0px 5px 5px 5px;
}

#navigation {


<! Question 3.1.2     >

#rightside {
position:absolute;left:320px; right:20px;width: 150px;
padding: 5px 15px;
}

<! Question     3.1.2 >

a { text-decoration: none;  }

<! Question     3.1.3>
a:hover  {background-color:gray;  }
```

```
<!Question      3.1.4>
#footer {
clear: both;
background-color:silver;
text-align:center;

}

<! Question 3.2  3.2.1>
<html>
<head>
<title>WBE Survey</title>
<link rel="stylesheet"href="style1.css"type="text/css">
</head>
<body>
<div id="container">
 <div id="header">
          <! Question 3.2.2     >
          <h1>  WBE SURVEY FOR TVET COLLEGES!</h1>
   </div>


<! Question     3.2.3>
<div id="navigation">
     <a href="">TVET Colleges</a>
     <a href="">Sponsorship </a>
     <a href="">Application form</a>
 </div>


<! Question 3.2.4>
<div id="rightside">
<img src="images/survey2.jpg
 "height="150px" width="230px"  >
</div>


<! Question 3.2.5>
<div id="content">
<table border="0">
<form action="" method="post" onsubmit="return ()">

<! Question     3.2.6>
<tr><td>  Surname:</td> <td><input type="text" name="surname1"
id="sname1" /></td></tr>
<tr><td>Email: </td> <td>
<input type="text" name="email" id="email" /></td></tr>
<tr><td> Specialization:<br></td> <td>
<INPUT TYPE="radio" NAME="rad" Value="Networking" >Networking <BR>
<INPUT TYPE="radio" NAME="rad" Value="Programming" >Programming<BR>
<INPUT TYPE="radio" NAME="rad" Value="System
Analysis">Programming<BR></td></tr>
```

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```
<! Question     3.2.7>
<span id="cty">
<tr><td>Province:
</td> <td><select name="province"
id="province">
<option value="">Choose a province</option>
<option value="ec">Eastern Cape</option>
<option value="gp">Gauteng</option>
<option value="fs">Free State</option>
<option value="kzn">Kwazulu Natal</option>
<option value="lp">Limpopo</option>
<option value="mp">Mpumalanga</option>
<option value="nc">Northern Cape</option>
<option value="nw">North West</option>
<option value="wc">Western Cape</option></td></tr>
</select>


<! Question 3.2.8>

</span><br><br>
<tr><td>Your Comment:</td><td><TEXTAREA NAME="myarea" COLS="40" ROWS="5">
</TEXTAREA></td></tr>
<tr><td></td><td><input type="submit"
value="Submit" />
   
<input type="reset"
value="Clear" /></td></tr>
</table></form>
</div>


<! Question     3.2.9>
<div id="footer">
        YOUR EXAMINATION MUMBER
    </div>
</div>
</body
</html>
```

QUESTION 4

```
<! Question 4.1                         4.1.1>
<script type="text/javascript">

<! Question 4.1.2>
function survey()
{
var txt = document.getElementById("sname1");
var txte = document.getElementById("email");
```

```
if(txt.value== "" || txt.value== null)
 {
alert("Please enter a name");
txt.style.border = "2px solid red";
txt.focus()
return false;
} else

{
txt.style.border = "";
}
if(txte.value== "" || txte.value== null)
 {
alert("Please enter an email");
txte.style.border = "2px solid red";
txte.focus()
return false;
} else

{
txte.style.border = "";
}

<! Allocate 4 to each text box validation >

<! Question 4.1.3>
var combo1 = document.getElementById("province")
if   (combo1.value == null || combo1.value == "")
 {
 )

<! Question                                    4.1.4>
                                               alert("Please select a province");
document.getElementById("cty").style.border = "2px solid red";
return false;
} else
{
document.getElementById("cty").style.border = "";
}
}
</script>
```

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